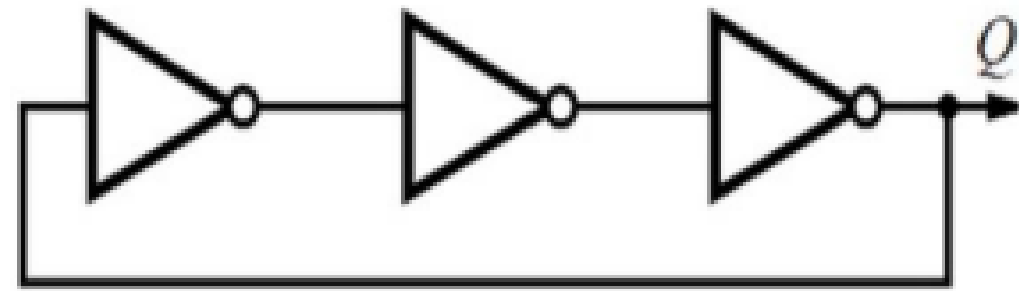


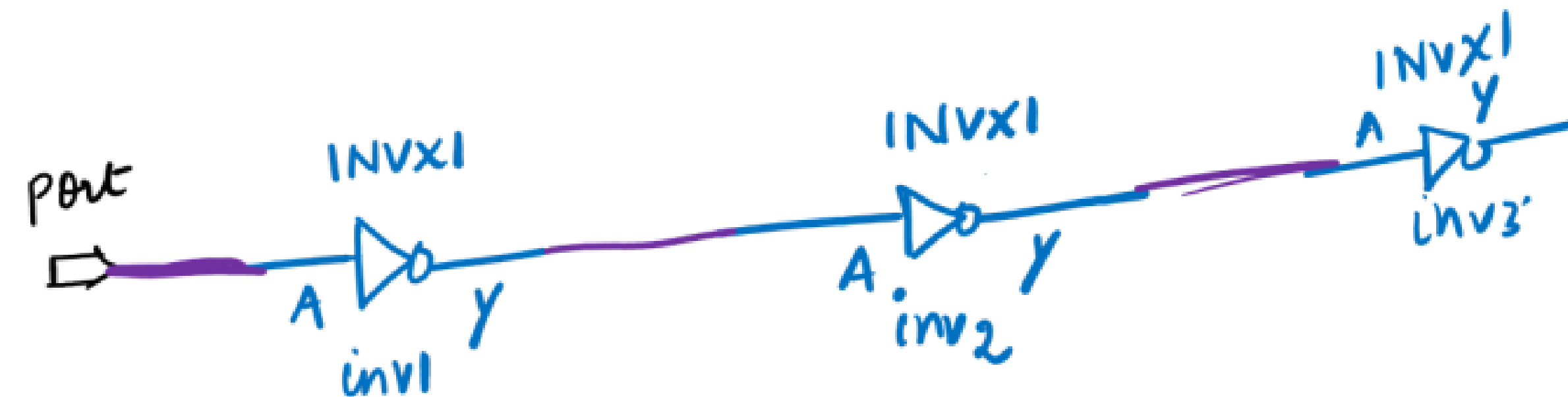
Design a ring oscillator using TcL (asked in 1 company written test round)



Verilog file explanation:

```

1 create_port port -direction inout
2
3
4 set prev_net port
5 set num_inverters 3
6
7 for {set i 1} {$i <= $num_inverters} {incr i} {
8     create_instance inv$i -of_module INVX1
9     create_connection $prev_net inv$i/A
10    set prev_net inv$i/Y
11    puts $prev_net
12 }
13 create_connection $prev_net port
  
```



1st iteration:

$i = 1$ $\$i \leq \$num_inverters$

'\$' in TCL $\Rightarrow$ content of var

$1 \leq 3 \checkmark \rightarrow$ Go into loop

* $prev_net \leftarrow port$

$\$prev_net \leftarrow port$

* updating $prev_net$ variable.
 $prev_net \leftarrow inv1/Y$

2nd iteration

$i = 2$

$2 \leq 3 \checkmark$

$\rightarrow$ Go into loop

$prev_net \leftarrow inv1/Y$
 $\$prev_net \leftarrow inv1/Y$

create_connection $\$prev_net$ inv*i*/A
inv1/Y inv2/A

* updating $prev_net$ variable
 $prev_net \leftarrow inv2/Y$

3rd iteration

$i = 3$

$3 \leq 3 \checkmark$

$\rightarrow$ Go into loop

$prev_net \leftarrow inv2/Y$
 $\$prev_net = inv2/Y$

create_conn $\$prev_net$ inv*i*/A
inv2/Y inv3/A

* updating $prev_net$ variable
 $prev_net \leftarrow inv3/Y$

4th iteration

$i = 4$

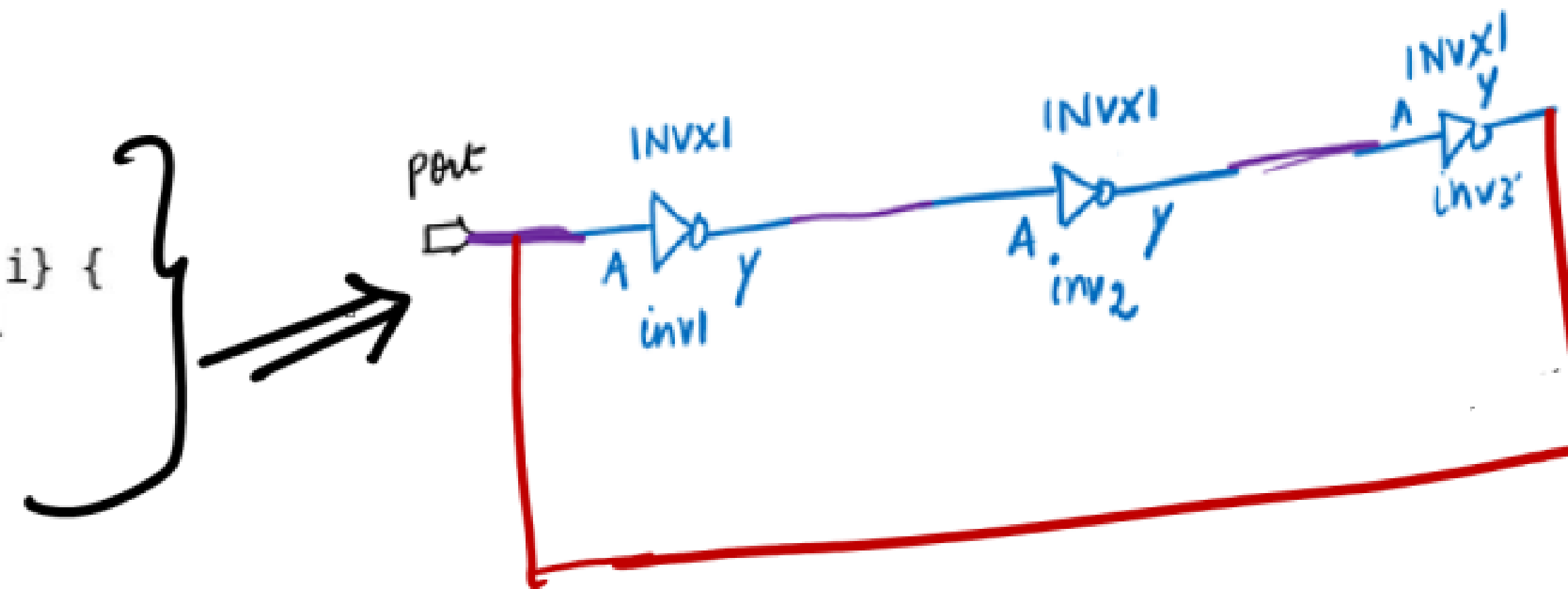
$4 \leq 3 \times$

$\rightarrow$ come out of loop


```

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12 }
13 create_connection $prev_net port

```



outside the loop

$\$prev_net \leftarrow inv3/Y$

↑
This gets connected to port

How to execute the TCL script :

```
1 set_context dft -scan  
2 read_verilog try.v ← empty file with just module and endmodule  
3 read_cell_library /hdd2/home/vidishar/nov24/library/tsmc13.mdt  
4 set_current_design  
5 set_sys m insertion  
6 source ring.tcl
```

7 o/p:

